

DETAGNON BERNIC GBAGUIDI

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PROFILE

PhD student in Mathematical Modeling at Rochester Institute of Technology, with prior training in pure mathematics, statistics for life sciences, and mathematical sciences. Current research on **category theory for machine learning**; broader interests span **machine learning, statistical modeling, and mathematical models for infectious disease and public health**, with a focus on systems that hold up under real-world constraints such as cost, latency, generalizability, and interpretability. Comfortable moving from mathematical formulation to production deployment.

EDUCATION

Aug 2025 – Present	PhD, Mathematical Modeling <i>Rochester Institute of Technology (RIT), New York, USA</i> Advisor: Dr. Ernest Fokoué. Member of the Data Science Research Group.
Sep 2023 – Oct 2024	MSc, Mathematical Sciences (Malaria Modeling) <i>African Institute for Mathematical Sciences (AIMS), Kigali, Rwanda</i> Thesis: <i>Retrospective Impact Analysis of Seasonal Malaria Chemoprevention in Benin Republic</i> . Evaluated the effectiveness of SMC campaigns in Benin using interrupted time series analysis on national surveillance data.
Oct 2022 – Jul 2023	Master 1 (M1), Statistics for Life Sciences <i>International Chair of Mathematics and Physics Applied (ICMPA), Abomey-Calavi, Benin</i>
Nov 2017 – Jul 2021	BSc, Pure Mathematics <i>University of Abomey-Calavi, Benin</i>

PUBLICATIONS

- Friederike Hartkopf, Alexis Robitaille, **Detagnon Bernic Gbaguidi**, et al. *Long-read sequencing reveals HIV-1 envelope correlates of autologous neutralizing antibody responses*. Under review, *EBioMedicine*.
- Gbaguidi, D. B.**, et al. *Digital Traces of Community Discourse: Unsupervised Topic Modeling of X/Twitter Posts in Rochester, NY*. In preparation.

TALKS

Oct 2026	Categorical Foundations of Machine Learning: An Introduction <i>Data Science Research Group Seminar, School of Mathematics and Statistics, RIT</i>
Oct 2024	Student Speaker, 9th AIMS Rwanda Graduation Ceremony <i>African Institute for Mathematical Sciences, Kigali, Rwanda</i>

RESEARCH EXPERIENCE

Apr 2025 – Aug 2025	Research Intern, Machine Learning for Infectious Disease <i>University of Tübingen, Germany</i> • Supervisor: Prof. Dr. Nico Pfeifer - Applied TabPFN, a transformer-based prior-fitted network for tabular data, to predict HIV-1 autologous neutralization status from features derived from viral envelope protein sequences. - Built the modeling pipeline on long-read sequencing data of HIV-1 envelope, from feature engineering through evaluation. - Contributed to the analysis and manuscript now under review at <i>EBioMedicine</i> (third author).
May 2024 – Sep 2024	Research Intern, National Malaria Control Program (PNLP) <i>Ministry of Health, Cotonou, Benin</i> - Conducted the applied component of the AIMS MSc thesis on Seasonal Malaria Chemoprevention (SMC) effectiveness in Benin. - Worked with national malaria surveillance data to design and estimate an interrupted time series model quantifying SMC impact.

SELECTED PROJECTS

- Mathiis, AI Math Exam Generator** (mathiis.com, in production). End-to-end system generating APC-format math exams for the Beninese secondary curriculum (6^{ème} to Terminale): Playwright scraper, Docling PDF extraction, ChromaDB vector store

with mxbai-embed-large-v1 embeddings, LangChain RAG pipeline, Gradio interface on Hugging Face Spaces. Iterating on a latency architecture (generator + judge, parallelized planning) targeting sub-minute generation under a tight per-exam cost budget.

- **Agrivoltaic Crop Placement Modeling (Alaska).** Full MATLAB pipeline modeling spinach yield under solar panels at a high-latitude site: corrected shading geometry, bolting submodels, yield optimization.
- **Civic Twitter Discourse in Rochester.** Research paper (see Publications). City-level analysis structured around cross-model convergence.
- **Virtual Try-On MVP for West African Clothing Sellers.** Hugging Face Spaces deployment using CatVTON with a WhatsApp-native product-link flow, targeted at small clothing retailers in the West African market.

TEACHING

Sep 2025 – Present	Graduate Teaching Assistant <i>Rochester Institute of Technology (RIT), New York, USA</i> • Support instruction for Probability & Statistics and Calculus courses. • Grade assignments and exams; hold weekly office hours to work with students on problem sets and concepts.
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TECHNICAL SKILLS

Programming	Python, R, MATLAB, C++, C
Machine Learning	LangChain, Hugging Face (Transformers, Spaces), ChromaDB, Gradio, embedding models, retrieval-augmented generation, LLM API workflows
Modeling & Statistics	Interrupted time series analysis, Monte Carlo simulation, mathematical modeling and optimization, statistical modeling for life sciences and epidemiology
Data Engineering	Web scraping (Playwright), document extraction (Docling), PDF pipelines (ReportLab, WeasyPrint, pdflatex)
Tools	Git, LaTeX, MS Office, LibreOffice
Languages	French (native), English (fluent, professional working proficiency)

AWARDS & SCHOLARSHIPS

Oct 2024	AIMS Exceptional Excellence Award <i>In recognition of an outstanding embodiment of the AIMS core value of excellence.</i>
2023 – 2024	AIMS Full Scholarship <i>Full merit-based scholarship for MSc studies at AIMS Kigali.</i>

REFERENCES

Available upon request.